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# Token-guided multimodal prognosis in hepatocellular carcinoma: a framework steered by tumour–stroma ratio

He-Yu Huang,<sup>1,2</sup> Kun Wu,<sup>3</sup> Li-mei Qu,<sup>4</sup> Xiao-Dong Sun,<sup>1,2</sup> Ming-Yue Li,<sup>1,2</sup> Feng Wei ,<sup>1,2</sup> Ping Zhang,<sup>1,2</sup> Alfred Wei Chieh Kow,<sup>5</sup> Yu-Guo Chen,<sup>1,2</sup> Mei-Shan Jin,<sup>4</sup> Liang Guo,<sup>4</sup> Wei Qiu,<sup>1,2</sup> Meng Wang,<sup>1,2</sup> Xiao-Ju Shi,<sup>1,2</sup> Jun-Feng Ye,<sup>1,2</sup> Chuan-Hao Hu,<sup>3</sup> Yue-Xuan Zhao,<sup>1,2</sup> Yu Huang,<sup>1,2</sup> Zhong-Qi Fan,<sup>1,2</sup> Yu-Shan Zheng,<sup>6</sup> Feng-Ying Xie,<sup>3</sup> Guo-Yue Lv ,<sup>1,2</sup>

► Additional supplemental material is published online only. To view, please visit the journal online (<https://doi.org/10.1136/gutjnl-2025-337945>).

For numbered affiliations see end of article.

## Correspondence to

Guo-Yue Lv; lvgy@jlu.edu.cn  
Professor Zhong-Qi Fan;  
fanzhongqi@jlu.edu.cn  
Professor Yu-Shan Zheng;  
yszheng@buaa.edu.cn  
and Professor Feng-Ying Xie; xfy\_73@buaa.edu.cn

HH, KW and LQ are joint first authors.

Received 24 December 2025  
Accepted 24 February 2026



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**To cite:** Huang H-Y, Wu K, Qu L, et al. *Gut* Epub ahead of print: [please include Day Month Year]. doi:10.1136/gutjnl-2025-337945

## ABSTRACT

**Background** The tumour–stroma ratio (TSR) is a potential prognostic indicator, yet hindered by quantification challenges and conflicting reports.

**Objective** To determine whether TSR follows a non-linear prognostic pattern and to develop an artificial intelligence (AI)-powered framework for standardised TSR assessment and prognosis prediction in hepatocellular carcinoma (HCC).

**Design** We integrated whole-slide image (WSI) data with clinical variables across a retrospective cohort (n=392) and The Cancer Genome Atlas dataset (n=168). Restricted cubic splines were used to interrogate non-linear hazard dynamics, with biological validation via transcriptomics and immunohistochemistry. An AI-driven foundation model framework was developed for TSR quantification and multimodal prognostic modelling.

**Results** Our analysis unveiled an inverted U-shaped non-linear relationship between TSR and mortality, identifying a risk initiation threshold at 0.188 and a peak at 0.268. Transcriptomics analysis indicated that this high-risk phenotype is characterised by active tumour proliferation, stromal activation and tumour microenvironment crosstalk. Technically, AI-derived TSR showed strong correlation with expert assessment ( $R^2 > 0.9$ ). Furthermore, we developed a novel 'Token-Guided Multimodal Fusion' architecture to integrate WSI, TSR and clinical variables as high-dimensional tokens directly into the computational logic. Consequently, our multimodal framework demonstrated prognostic accuracy (area under the curve  $> 0.80$ ) compared with unimodal baselines.

**Conclusion** This study redefines TSR assessment, shifting from manual estimation to high-dimensional semantic reasoning. By identifying the non-linear prognostic mechanics of the stroma, our token-guided framework offers a biologically interpretable solution for HCC. We suggest that the future of computational pathology may lie not in simple quantification, but in the semantic fusion of human domain knowledge with AI reasoning.

## WHAT IS ALREADY KNOWN ABOUT THIS TOPIC

- ⇒ The tumour–stroma ratio (TSR) is a promising prognostic factor in solid tumours, yet its application is limited by quantification challenges and conflicting prognostic reports.
- ⇒ Current assessments typically rely on manual estimation or supervised segmentation artificial intelligence (AI), which treats TSR as a static scalar and assumes a linear relationship with prognosis.

## WHAT THIS STUDY ADDS

- ⇒ Non-linear discovery: This study identified a distinct inverted U-shaped relationship between TSR and hepatocellular carcinoma (HCC) prognosis, defining a specific risk window that challenges the traditional linear assumption and explaining previous contradictory reports.
- ⇒ Methodological innovation: We developed a foundation model (Conch v1.5)-driven framework that enables precise, annotation-free TSR quantification, showing strong correlation with expert pathologists' delineation.
- ⇒ Architectural innovation: We developed a novel 'Token-Guided Multimodal Framework' that transforms TSR from a simple scalar into a high-dimensional 'semantic anchor' alongside whole-slide image features. This design explicitly recalibrates the model's attention to stromal components, correcting cellularity bias and achieving superior prognostic accuracy with robust cross-cohort generalisability. Notably, our model autonomously resolved the inverted U-shaped prognostic relationship without explicit supervision.

## INTRODUCTION

Despite advances in systemic therapy, hepatocellular carcinoma (HCC) remains characterised by marked prognostic heterogeneity that existing staging systems fail to capture.<sup>1–3</sup> While the tumour micro-environment (TME) is recognised as a key determinant of tumour progression, current research

## HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

- ⇒ It provides a standardised, automated tool for precise TSR assessment, enabling consistent risk stratification focusing on stroma across clinical centres to complement current HCC staging.
- ⇒ This study supports a shift in clinical modelling: moving from numerical-based predictions to high-dimensional token-based reasoning.
- ⇒ It establishes a promising example of 'Human-Guided AI', demonstrating that integrating domain knowledge (TSR logic) into deep learning architectures can correct algorithmic biases and enhance the biological interpretability of clinical prediction models.

has disproportionately focused on immune landscapes, leaving structural pathological parameters largely unexplored.<sup>4–8</sup> Among these, the tumour–stroma ratio (TSR) has emerged as a potent prognostic surrogate in solid tumours like colorectal and breast cancer.<sup>9–12</sup> However, its application in HCC remains contentious and underexplored.<sup>13</sup>

Unlike the desmoplasia-rich stroma of pancreatobiliary or breast cancers,<sup>14</sup> HCC exhibits a unique stromal architecture characterised by scant stromal components, varying degrees of sinusoidal capillarisation and distinct cancer-associated fibroblast subpopulations.<sup>15</sup> This subtle and variable landscape renders conventional visual estimation inadequate. Furthermore, current TSR assessments typically rely on single-section manual inspection.<sup>14–16</sup> This approach is not only labour-intensive and subject to interobserver variability but also fails to capture the spatial heterogeneity inherent to the liver as the largest solid organ, thereby compromising prognostic reliability.

The advent of artificial intelligence (AI) offers a pathway to overcome these limitations. Although several studies have applied AI to automate grading and structural feature recognition (such as microvascular invasion, MVI),<sup>17–19</sup> no study has yet integrated spatially representative sampling with expert-guided precision to decode the prognostic value of TSR in HCC. Moreover, current AI approaches treat TSR as simple pixel ratios, ignoring the high-dimensional biological information embedded in stromal morphology. Foundation models offer the potential to bridge this gap, yet their application to TSR remains unexplored.

To address these gaps, this study pursued two interrelated objectives. First, we sought to determine whether the TSR–prognosis relationship follows a non-linear pattern and to identify biologically meaningful thresholds for risk stratification. Building on this, we aimed to develop an AI-powered framework for standardised and automatic, annotation-free TSR quantification and multimodal prognosis prediction, with traditional Cox regression models serving as baseline comparators. By integrating TSR as a high-dimensional token within a multimodal architecture, we aim to enable a more informative representation of TSR for risk stratification and prognosis modelling in HCC.

## METHODS

### Study cohorts from JLU and TCGA

We retrospectively analysed 508 consecutive patients with HCC who underwent curative hepatectomy at the First Hospital of Jilin University (JLU cohort) between January

2017 and December 2021. Patients were excluded if they had: (1) non-HCC or mixed histological subtypes (n=64); (2) prior locoregional or systemic therapy (n=25); (3) incomplete clinicopathological or follow-up data (n=12); or (4) poor-quality histological slides unsuitable for digital pathology analysis (n=15). After applying these criteria, 392 patients were enrolled for model development. The median follow-up time was 35.92 months (IQR, 23.63–51.03). Overall survival (OS) was defined as the primary endpoint. Tumour, node, metastasis (TNM) stage was assigned according to the American Joint Committee on Cancer (AJCC) eighth edition staging system. For external validation, 168 patients with HCC were selected from The Cancer Genome Atlas Liver Hepatocellular Carcinoma (TCGA-LIHC) cohort and matched to the JLU cohort by age, sex and AJCC TNM stage (figure 1A). Baseline clinicopathological variables are summarised in table 1.

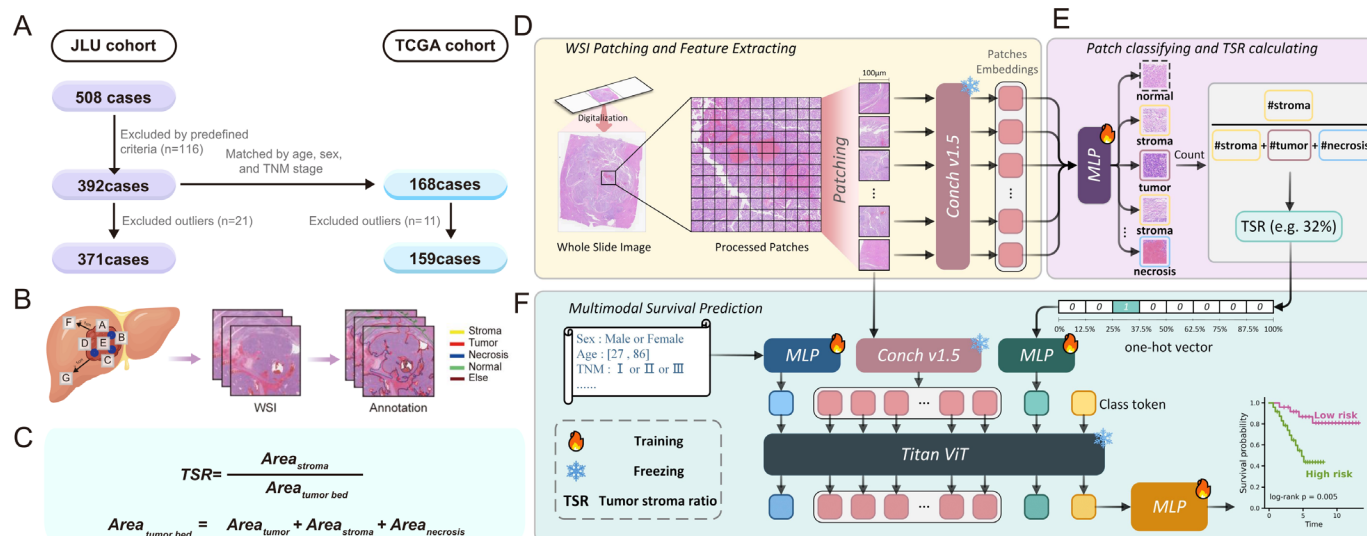
### Whole-slide imaging acquisition and pathologist annotation

Given the large tumour volume and spatial heterogeneity characteristic of HCC, tissue sampling in the JLU cohort followed a standardised seven-point protocol in accordance with routine clinical practice. Sections of inadequate quality (eg, extensive necrosis) were excluded. An experienced hepatopathologist selected one representative section that best captured the overall tumour–stroma composition, which was digitised at 40× magnification using a Motic scanner and converted to SVS format. Digital slides were reviewed and annotated using ImageScope software at 10× magnification. Five tissue categories (tumour, stroma, necrosis, normal liver, background) were manually delineated by two hepatopathologists (figure 1B). Clinician-assessed TSR (cTSR) was calculated as the proportion of stromal area within the tumour bed, where the tumour bed comprised tumour cells, stromal components and necrotic areas, within the annotated regions (figure 1C). This definition served as the gold standard for TSR quantification and model training. Regions corresponding to empty background or obvious artefacts were excluded.

### AI-based patch classification and TSR prediction

An overview of the training status, data sources, validation strategy and evaluation metrics for each component of the multimodal framework is provided (online supplemental Table S1). As a core component, we developed an AI-based pipeline for patch-level tissue classification and automated TSR prediction. Whole-slide images (WSIs) were preprocessed to remove background and non-tissue regions and tiled into non-overlapping 256×256-pixel patches at 20× magnification. The distribution of the number of patches per WSI in the JLU and TCGA cohorts is summarised (online supplemental Table S2). Patch-level embeddings (768-dimensional) were extracted using the pretrained pathology foundation model Conch v1.5.<sup>20</sup> An HCC-specific multilayer perceptron (MLP) classifier was trained to categorise each patch into tumour, stroma or necrosis based on CONCH features (figure 1D).

Predicted TSR (pTSR) was defined as a fully automated, whole-slide metric, calculated using patch-level tissue classification without manual annotation. During inference, patch classification was performed across the entire WSI; however, normal liver/background patches were explicitly identified as a separate class and excluded from TSR



**Figure 1** Study design and AI-based multimodal framework for TSR-guided survival prediction. (A) Patient selection and cohort construction in the JLU and TCGA cohorts. The JLU cohort was collected and included 392 patients for AI model development, with 371 retained for statistical analyses after outlier exclusion. The TCGA cohort served as an independent external validation set, with 159 of 168 patients included after outlier exclusion. (B) Representative section selection and annotation. Tumour sampling followed a standardised seven-point protocol, and digitised WSIs were annotated into predefined tissue categories. (C) Definition of the TSR. TSR was calculated as stromal area divided by the tumour bed area (tumour+stroma + necrosis). (D) WSI were tiled into non-overlapping patches, and patch-level features were extracted for downstream analysis. (E) TSR was computed based on the proportions of tissue categories predicted by the patch-level classification network. (F) Patch-level features, TSR representations and clinical variables were integrated into a multimodal prognostic network for end-to-end survival prediction. AI, artificial intelligence; JLU, First Hospital of Jilin University; MLP, multilayer perceptron; TCGA, The Cancer Genome Atlas; TNM, tumour, node, metastasis; TSR, tumour–stroma ratio; ViT, Vision Transformer; WSI, whole-slide image.

computation. pTSR was calculated as the ratio of stromal patches to the total number of tumour, stromal and necrotic patches within tumour-containing regions (figure 1E). pTSR was used as a key quantitative variable for subsequent prognostic analyses.

### Development, validation and interpretability of the multimodal survival model

The multimodal survival model comprised token encoding, Vision Transformer (ViT)-based multimodal fusion<sup>21</sup> and discrete-time survival prediction.<sup>22</sup> Patch embeddings, TSR representations and clinical variables were encoded into learnable tokens and jointly integrated into a ViT-based survival prediction network built on the pretrained Titan foundation model<sup>21</sup> (figure 1F) to capture cross-modal interactions. Clinical variables were transformed into normalised or discretised representations according to predefined encoding rules before integration (online supplemental Table S3). The network performed multimodal fusion to generate patient-level representations for end-to-end prognostic estimation.

The follow-up period was divided into four non-overlapping risk intervals for discrete-time survival prediction. Model performance was assessed as described in the Statistical Analysis section using time-dependent area under the receiver operating characteristic (AUC)<sup>23</sup> at 60 months (with inverse probability of censoring weighting) and the concordance index (C-index). Model interpretability was evaluated using SHapley Additive exPlanations (SHAP),<sup>24</sup> and feature contributions were visualised using beeswarm plots. To further examine potential non-linear effects at the modelling level, patients were stratified into restricted cubic spline (RCS)-defined TSR subgroups, and models

were trained and evaluated within individual subgroups and predefined subgroup combinations using identical architectures and training protocols. In addition, representative low-TSR and high-TSR cases were selected to generate WSI-level attention heatmaps for deep learning (DL)<sub>WSI</sub> and DL<sub>WSI+TSR</sub> models, enabling qualitative comparison of spatial attention patterns following TSR integration.

### Statistical analysis

#### Survival analysis

OS was defined as the time from curative hepatectomy to death from any cause or last follow-up. Patients alive at the last follow-up were treated as censored observations. Survival distributions were estimated using the Kaplan-Meier method and compared using the log-rank test. To test our hypothesis, associations between TSR and OS were evaluated using Cox proportional hazards regression models. The proportional hazards assumption was assessed using Schoenfeld residuals. Univariable analyses were followed by multivariable Cox models adjusting for clinicopathological covariates; in the TCGA cohort, multivariable analyses were performed using corresponding available variables.

#### Restricted cubic spline modelling

RCS functions were applied within the Cox framework to explore potential non-linear associations between TSR and OS. RCS models were fitted using four knots placed at the 5th, 35th, 65th and 95th percentiles of the TSR distribution. RCS analyses were used to characterise the shape of the TSR–survival relationship and to identify biologically meaningful inflection points for subsequent risk stratification.<sup>25</sup>



**Table 1** Baseline clinicopathological characteristics of the study cohorts (JLU and TCGA)

| Variable                 | JLU cohort (n=392 )<br>cases (%) | TCGA cohort (n=168 )<br>cases (%) | P value |
|--------------------------|----------------------------------|-----------------------------------|---------|
| cTSR                     | 0.18 (0.11–0.27)                 | 0.16 (0.08–0.26)                  | 0.2489  |
| Age                      | 58.00 (51.00–65.00)              | 59.00 (53.75–66.00)               | 0.0590  |
| Sex                      |                                  |                                   | 0.9743  |
| Male                     | 304 (77.6)                       | 132 (78.6)                        |         |
| Female                   | 88 (22.4)                        | 36 (21.4)                         |         |
| TNM stage                |                                  |                                   | 0.2427  |
| I                        | 209 (53.3)                       | 97 (57.7)                         |         |
| II                       | 137 (34.9)                       | 44 (26.2)                         |         |
| III                      | 46 (11.7)                        | 27 (16.1)                         |         |
| AFP (ng/mL)              |                                  |                                   | 0.6524  |
| <400                     | 306 (78.1)                       | 143 (85.1)                        |         |
| ≥400                     | 86 (21.9)                        | 25 (14.9)                         |         |
| Largest tumour size (cm) |                                  | NA                                | NA      |
| <5                       | 261 (66.6)                       |                                   |         |
| ≥5                       | 131 (33.4)                       |                                   |         |
| Tumour capsule           |                                  | NA                                | NA      |
| Present                  | 140 (35.7)                       |                                   |         |
| Absent                   | 252 (64.3)                       |                                   |         |
| MVI                      |                                  | NA                                | NA      |
| M0                       | 200 (51.0)                       |                                   |         |
| M1                       | 127 (32.4)                       |                                   |         |
| M2                       | 65 (16.6)                        |                                   |         |
| HBV                      |                                  |                                   |         |
| No                       | 118 (30.1)                       |                                   |         |
| Yes                      | 274 (69.9)                       |                                   |         |
| HCV                      |                                  | NA                                | NA      |
| No                       | 341 (87.0)                       |                                   |         |
| Yes                      | 51 (13.0)                        |                                   |         |
| Cirrhosis                |                                  | NA                                | NA      |
| No                       | 178 (45.4)                       |                                   |         |
| Yes                      | 214 (54.6)                       |                                   |         |

Continuous variables are presented as median (IQR), and categorical variables as number (percentage). Continuous variables were analysed using either the Student's t-test or the Mann-Whitney U test, depending on data distribution, while categorical variables were compared using the  $\chi^2$  test or Fisher's exact test. AFP, alpha-fetoprotein; cTSR, clinician-assessed tumour–stroma ratio; JLU, First Hospital of Jilin University; MVI, microvascular invasion; NA, not assessed; TCGA, The Cancer Genome Atlas; TNM, tumour, node, metastasis staging.

### Model validation and performance evaluation

Discrimination was assessed using Harrell's C-index and time-dependent AUC, while calibration and clinical utility were evaluated using time-dependent calibration plots and decision curve analysis (DCA). These metrics were applied consistently to AI-based multimodal prognostic models and to conventional Cox regression models used as baseline comparators. In addition, ablation analyses were conducted by removing TSR from otherwise identical multimodal models to quantify the incremental prognostic value of TSR beyond clinical variables.

### Sensitivity analyses

Sensitivity analyses were performed to evaluate the robustness of the TSR–survival association under alternative model specifications and exclusion of extreme values. To evaluate potential confounding effects of necrosis, sensitivity analyses were performed by recalculating TSR using a conventional

formulation excluding necrotic regions (stroma/[stroma+tumour]) and repeating RCS analyses based on this definition.

### Subgroup analysis

Prespecified subgroup analyses were conducted according to clinically relevant variables, including sex, alpha-fetoprotein (AFP) level, MVI status and TNM stage. Interaction terms were tested to assess effect modification across subgroups. All statistical tests were two-sided, and  $p < 0.05$  was considered statistically significant.

### Software and statistical environment

Statistical analyses were conducted using Python (V3.13.11) with the lifelines package (V0.30.0) for survival analyses, scikit-learn (V1.7.1) for predictive modelling and statsmodels (V0.14.5) for advanced statistical analyses. Differential expression and pathway enrichment analyses were performed in R using DESeq2<sup>26</sup> and gene set enrichment analysis (GSEA).<sup>27</sup> DL models were implemented in Python (V3.10) using PyTorch (V2.5.1). All DL experiments were conducted on an NVIDIA GeForce RTX 4090 GPU.

### Patient and public involvement

Patients or the public were not involved in the design, conduct, reporting or dissemination plans of this research.

Other methods used in this study can be found in online supplemental methods.

## RESULTS

### Baseline clinical characteristics and TSR distribution of the study cohorts

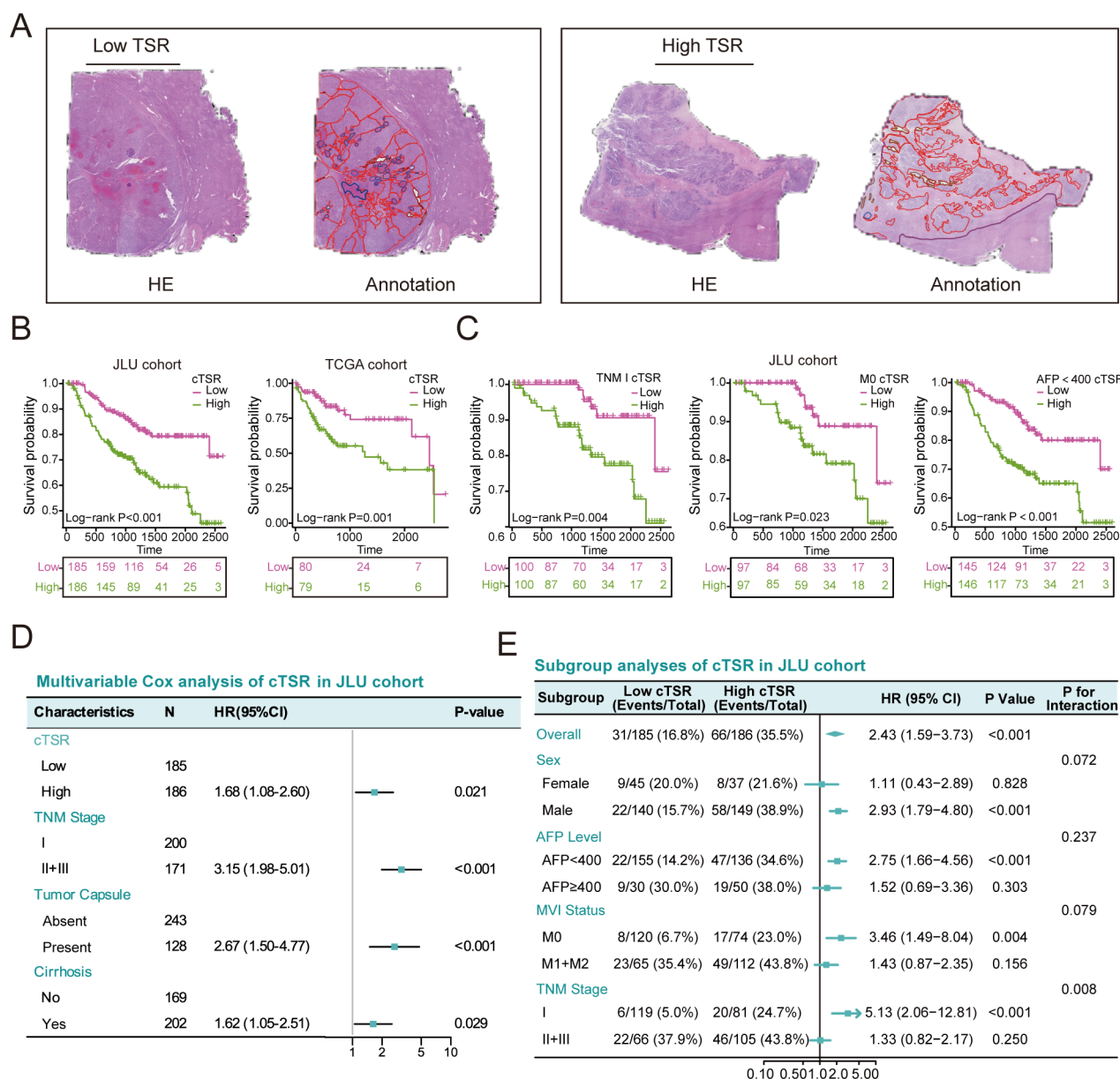
The JLU cohort comprised 392 patients with resected HCC (median age, 58 years (IQR, 51.0–65.0); 77.6% male). The median cTSR was 0.18 (IQR, 0.11–0.27). TNM stage I, II and III disease accounted for 53.3%, 34.9% and 11.7% of patients, respectively. Serum AFP levels were  $\geq 400$  ng/mL in 21.9%. Largest tumour size  $\geq 5$  cm was present in 33.4%, and MVI was observed in 49.0% (M1, 32.4%; M2, 16.6%). HBV and HCV infection was present in 69.9% and 13.0%.

The TCGA cohort included 168 patients with comparable baseline characteristics (median age, 59 years (IQR, 53.8–66.0); 77.6% male). The median cTSR was 0.16 (IQR, 0.08–0.26). TNM stage I–III disease accounted for 57.7%, 26.2% and 16.1%, respectively, and 14.9% had AFP levels  $\geq 400$  ng/mL. No statistically significant differences were observed between the JLU and TCGA cohorts with respect to cTSR, age, sex, TNM stage or AFP level (all  $p > 0.05$ ). The clinicopathological characteristics of both cohorts are summarised in [table 1](#).

### TSR is an independent prognostic factor in HCC

Representative histopathological examples illustrating low and high TSR, highlighting distinct stromal compositions within the TME, are shown in ([figure 2A](#)). Patients were stratified into low and high cTSR groups using the cohort-specific median as the cut-off. Kaplan-Meier analyses demonstrated that patients with high cTSR had significantly poorer OS in the JLU cohort (log-rank  $p < 0.001$ ; [figure 2B](#)). This finding was externally validated in the TCGA cohort (log-rank  $p = 0.001$ ; [figure 2B](#)). Subgroup Kaplan-Meier analyses showed that the adverse prognostic impact of high cTSR persisted in TNM stage I, MVI-negative (M0) status and AFP  $< 400$  ng/mL ([figure 2C](#)). These findings indicate that TSR retains prognostic value in early-stage HCC, where conventional risk markers are often limited. In





**Figure 2** Prognostic significance of TSR in HCC. (A) Representative histopathological images illustrating low-TSR and high-TSR cases. (B) Kaplan-Meier analyses of OS stratified by median TSR in the JLU cohort (n=371) and TCGA (n=159) cohort. Both cTSR and AI-predicted pTSR were significantly associated with OS (log-rank test). (C) Subgroup Kaplan-Meier analyses in the JLU cohort demonstrating the prognostic impact of cTSR in clinically relevant strata, including TNM stage I, microvascular invasion-negative (M0) and AFP <400 ng/mL subgroups (log-rank test). (D) Multivariable Cox proportional hazards analysis of cTSR and clinicopathological variables in the JLU cohort, with variables selected based on univariable Cox analyses (p<0.05). (E) Forest plot of prespecified subgroup analyses evaluating the consistency of the association between cTSR and OS across clinical subgroups. Interaction p values indicate effect modification. AFP, alpha-fetoprotein; AI, artificial intelligence; cTSR, clinician-assessed tumour–stroma ratio; HCC, hepatocellular carcinoma; JLU, First Hospital of Jilin University; MVI, microvascular invasion; OS, overall survival; pTSR, AI-predicted tumour–stroma ratio; TCGA, The Cancer Genome Atlas; TNM, tumour, node, metastasis; TSR, tumour–stroma ratio.

contrast, survival differences were less pronounced in advanced-stage subgroups (TNM stage II–III, MVI-positive (M1–2) and AFP≥400 ng/mL; online supplemental Fig. S1A).

In multivariable Cox proportional hazards regression analyses adjusted for age, sex, AFP level, TNM stage, cirrhosis, largest tumour size, tumour capsule, MVI and HBV status, high cTSR remained an independent predictor of mortality in the JLU cohort (HR 1.68, 95% CI 1.08 to 2.60; p=0.021) and was consistently validated in the TCGA cohort (HR 2.22, 95% CI 1.19 to 4.17; p=0.013; figure 2D; table 2; online supplemental Fig. S1B), supporting the robustness and cross-cohort generalisability of

cTSR-based prognostic stratification. Effect sizes are reported as HRs with 95% CIs, with multivariable estimates summarised in table 2 and validated in online supplemental Figure S1B, and subgroup and interaction effects presented in figure 2E.

Prespecified interaction analyses revealed significant effect modification by TNM stage (P for interaction=0.008), with a markedly stronger prognostic effect of high cTSR observed in TNM stage I patients (HR 5.13, 95% CI 2.06 to 12.81). No statistically significant interactions were observed for MVI status, sex or AFP level (all P for interaction >0.05) (figure 2E).

**Table 2** Univariable and multivariable Cox regression analyses of cTSR and clinicopathological factors for overall survival in the JLU and TCGA cohorts

| Variable            | JLU cohort (n=371) |                  |         |               |                  |         | TCGA cohort (n=159) |                  |         |               |                |         |
|---------------------|--------------------|------------------|---------|---------------|------------------|---------|---------------------|------------------|---------|---------------|----------------|---------|
|                     | Univariable        |                  |         | Multivariable |                  |         | Univariable         |                  |         | Multivariable |                |         |
|                     | HR                 | 95% CI           | P value | HR            | 95% CI           | P value | HR                  | 95% CI           | P value | HR            | 95% CI         | P value |
| cTSR                |                    |                  |         |               |                  |         |                     |                  |         |               |                |         |
| Low                 | –                  | –                | –       | –             | –                | –       | –                   | –                | –       | –             | –              | –       |
| High                | 2.454              | (1.601 to 3.762) | <0.001  | 1.675         | (1.081 to 2.596) | 0.021   | 2.477               | (1.389 to 4.418) | 0.002   | 2.223         | 1.187 to 4.166 | 0.013   |
| TNM stage           |                    |                  |         |               |                  |         |                     |                  |         |               |                |         |
| I                   | –                  | –                | –       | –             | –                | –       | –                   | –                | –       | –             | –              | –       |
| II+III              | 4.149              | (2.641 to 6.516) | <0.001  | 3.151         | (1.983 to 5.008) | <0.001  | 2.583               | (1.484 to 4.497) | <0.001  | 2.486         | 1.390 to 4.446 | 0.002   |
| AFP level           |                    |                  |         |               |                  |         |                     |                  |         |               |                |         |
| AFP <400 ng/mL      | –                  | –                | –       | –             | –                | –       | –                   | –                | –       | –             | –              | –       |
| AFP ≥400 ng/mL      | 1.554              | (1.001 to 2.413) | 0.049   |               |                  |         | 2.531               | (1.449 to 4.420) | 0.001   | 1.699         | 0.923 to 3.127 | 0.089   |
| Age                 | 0.999              | (0.980 to 1.019) | 0.951   |               |                  |         | 1.181               | (0.892 to 1.563) | 0.246   |               |                |         |
| Sex                 |                    |                  |         |               |                  |         |                     |                  |         |               |                |         |
| Female              | –                  | –                | –       | –             | –                | –       | –                   | –                | –       | –             | –              | –       |
| Male                | 1.467              | (0.868 to 2.477) | 0.152   |               |                  |         | 0.74                | (0.405 to 1.351) | 0.326   |               |                |         |
| Tumour capsule      |                    |                  |         |               |                  |         |                     |                  |         |               |                |         |
| Absent              | –                  | –                | –       | –             | –                | –       | –                   | –                | –       | –             | –              | –       |
| Present             | 0.263              | (0.149 to 0.463) | <0.001  | 2.675         | (1.499 to 4.773) | <0.001  |                     |                  |         |               |                |         |
| Cirrhosis           |                    |                  |         |               |                  |         |                     |                  |         |               |                |         |
| No                  | –                  | –                | –       | –             | –                | –       | –                   | –                | –       | –             | –              | –       |
| Yes                 | 2.002              | (1.301 to 3.080) | 0.002   | 1.625         | (1.052 to 2.512) | 0.029   |                     |                  |         |               |                |         |
| Largest tumour size |                    |                  |         |               |                  |         |                     |                  |         |               |                |         |
| < 5 cm              | –                  | –                | –       | –             | –                | –       | –                   | –                | –       | –             | –              | –       |
| ≥5 cm               | 3.462              | (2.310 to 5.189) | <0.001  |               |                  |         |                     |                  |         |               |                |         |
| MVI                 |                    |                  |         |               |                  |         |                     |                  |         |               |                |         |
| M0                  | –                  | –                | –       | –             | –                | –       | –                   | –                | –       | –             | –              | –       |
| M1+M2               | 4.129              | (2.613 to 6.524) | <0.001  |               |                  |         |                     |                  |         |               |                |         |
| HBV                 |                    |                  |         |               |                  |         |                     |                  |         |               |                |         |
| No                  | –                  | –                | –       | –             | –                | –       | –                   | –                | –       | –             | –              | –       |
| Yes                 | 1.884              | (1.141 to 3.111) | 0.013   |               |                  |         |                     |                  |         |               |                |         |
| HCV                 |                    |                  |         |               |                  |         |                     |                  |         |               |                |         |
| No                  | –                  | –                | –       | –             | –                | –       | –                   | –                | –       | –             | –              | –       |
| Yes                 | 0.57               | (0.287 to 1.133) | 0.109   |               |                  |         |                     |                  |         |               |                |         |

AFP, alpha-fetoprotein; cTSR, clinician-assessed tumour–stroma ratio; JLU, First Hospital of Jilin University; MVI, microvascular invasion; TCGA, The Cancer Genome Atlas; TNM, tumour, node, metastasis staging.

Given the attenuated effect in advanced-stage subgroups, we further explored the dose–response relationship between TSR and survival to determine whether alternative cut-offs might improve prognostic discrimination.

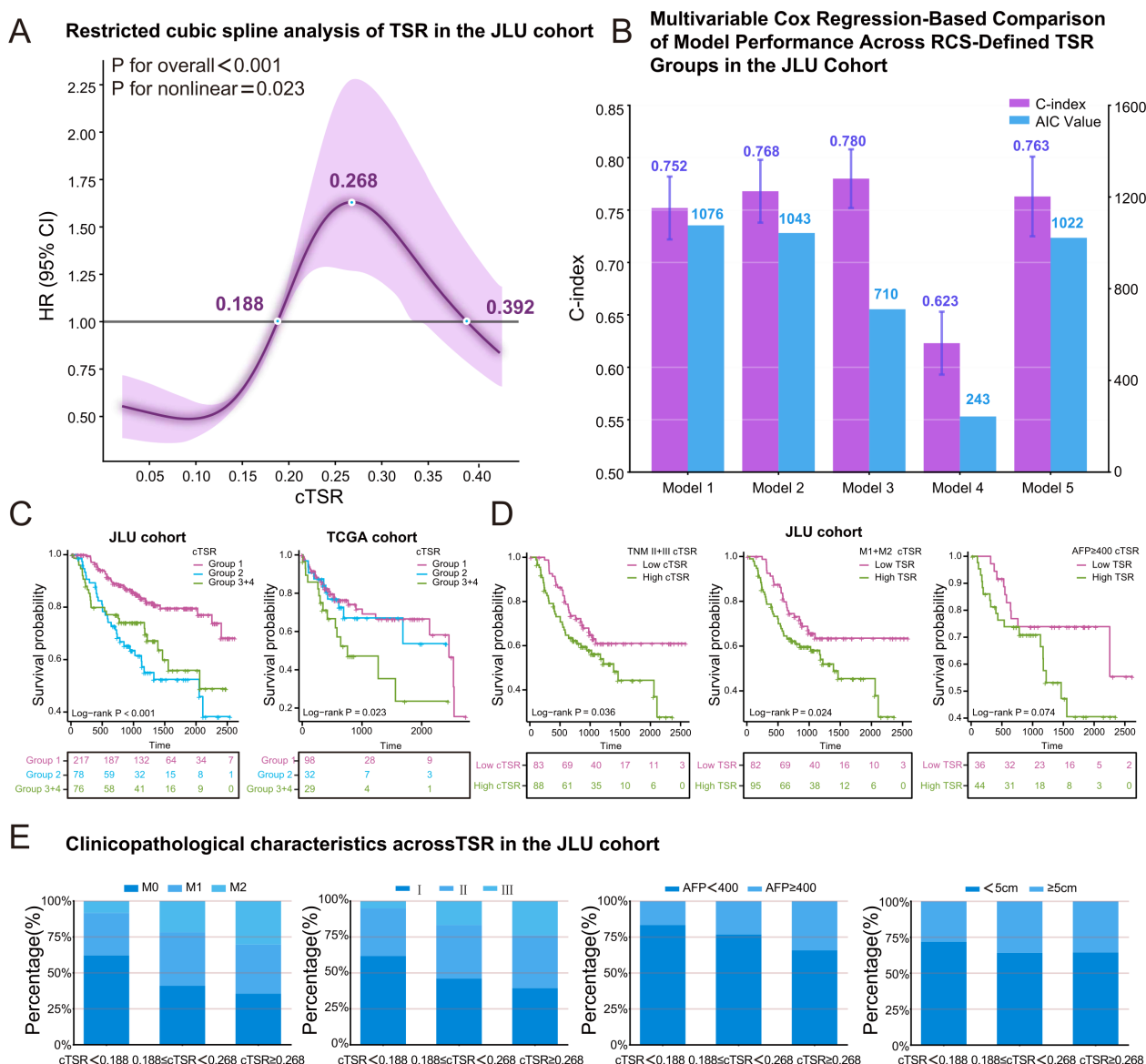
### Non-linear prognostic effect of TSR and RCS-guided risk stratification

RCS analyses revealed a significant non-linear association between cTSR and mortality ( $P$  for overall <0.001;  $P$  for non-linear=0.023; [figure 3A](#)). The HR increased progressively beyond 0.188, reached a plateau around 0.268 and declined in the ultra-high TSR range (>0.392), forming a distinct inverted U-shaped curve.

Comparison of Cox models showed that RCS-derived thresholds outperformed continuous and median-based representations, with higher C-indices and lower Akaike information criterion values ([figure 3B](#)). Using these thresholds, patients were stratified into risk groups.

Because the ultra-high TSR group (>0.392) contained only 10 patients, Groups 3 and 4 were combined to ensure statistical stability, yielding three final risk strata. Kaplan–Meier analyses demonstrated clear survival separation across these groups in both the JLU (log-rank  $p$ <0.001) and TCGA cohorts (log-rank  $p$ =0.023; [figure 3C](#)). When the RCS-derived 0.188 threshold was applied to advanced-stage subgroups, significant survival separation emerged in MVI-positive ( $p$ =0.024) and TNM stage II–III patients ( $p$ =0.036; [figure 3D](#)).

A similar trend was observed in the AFP ≥400 ng/mL subgroup ( $p$ =0.074), though limited by sample size ( $n$ =80). These findings indicate that RCS-guided cut-off selection extends cTSR-based risk stratification to advanced-stage populations. Clinicopathological composition varied across cTSR-defined strata, including TNM stage, MVI status, serum AFP level and tumour size ([figure 3E](#)).



**Figure 3** RCS analysis of TSR–survival association and RCS-guided risk stratification. (A) RCS curves illustrating the non-linear associations between cTSR and overall survival in the JLU cohort. Three inflection points were identified at 0.188, 0.268 and 0.392 ( $P$  for overall <0.001;  $P$  for non-linearity=0.023). (B) Comparison of Cox model performance using different cTSR representations (continuous, median-based and RCS-derived thresholds), evaluated by C-index and AIC. (C) Kaplan-Meier analyses of overall survival stratified by RCS-derived TSR risk groups in the JLU ( $n=371$ ) and TCGA ( $n=159$ ) cohorts (log-rank test). (D) Subgroup Kaplan-Meier analyses demonstrating survival separation using the RCS-derived 0.188 threshold in MVI-positive and TNM II–III subgroups (log-rank test). (E) Distribution of key clinicopathological characteristics across cTSR-defined strata, including AJCC TNM stage, MVI status, serum AFP level and tumour size. AFP, alpha-fetoprotein; AIC, Akaike information criterion; AJCC, American Joint Committee on Cancer; C-index, concordance index; cTSR, clinician-assessed tumour–stroma ratio; JLU, First Hospital of Jilin University; MVI, microvascular invasion; RCS, restricted cubic spline; TCGA, The Cancer Genome Atlas; TNM, tumour, node, metastasis; TSR, tumour–stroma ratio.

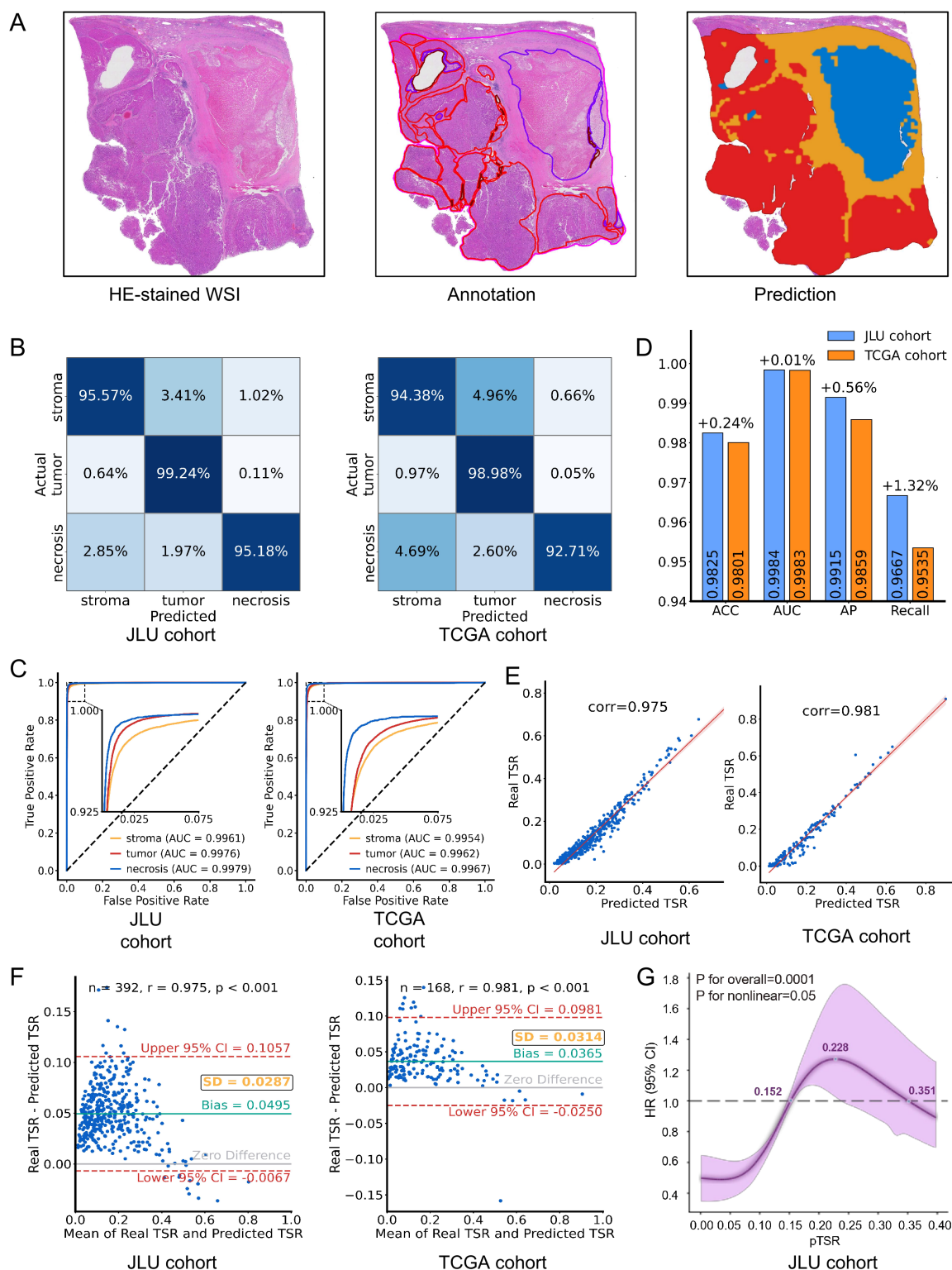
### Performance and reliability of patch-level classification and TSR estimation

Using the Conch v1.5 foundation model for feature extraction and an MLP classifier for tissue classification, WSIs from the JLU cohort yielded 386–17 563 valid tissue patches per slide (mean=2901), whereas TCGA slides contained 203–21 917 patches (mean=7630). The patch-level classification model accurately differentiated tumour, stroma and necrosis regions, as illustrated by representative segmentation examples (figure 4A).

High classification consistency was observed across tissue classes, with confusion matrices demonstrating robust agreement between predicted labels and pathologist annotations in both the JLU and TCGA cohorts (figure 4B). Class-wise receiver

operating characteristic (ROC) analyses showed strong discriminative performance for all tissue types, with AUC values consistently exceeding 0.95 (figure 4C) and overall classification metrics, including accuracy, AUC, average precision and recall, remained consistently high across cohorts (figure 4D). Furthermore, AI-predicted pTSR showed a strong linear correlation with cTSR, with Pearson correlation coefficients greater than 0.97 (figure 4E). Bland-Altman analyses demonstrated minimal systematic bias and narrow limits of agreement (figure 4F), indicating that AI-derived measurements achieve near-expert precision. Furthermore, RCS analysis of pTSR revealed a non-linear association with mortality similar to that of cTSR, with inflection points at 0.152, 0.228 and 0.351 ( $P$  for overall <0.001;





**Figure 4** Quantitative validation and survival relevance of pTSR. (A) Representative WSI regions illustrating automated tissue categorisation into tumour, stroma and necrosis, demonstrating consistency with routine histopathological structures. (B) Confusion matrices demonstrating robust agreement between model-predicted tissue labels and pathologist annotations in the JLU and TCGA cohorts. (C–D) ROC analysis and summary performance metrics, demonstrating robust and consistent tissue classification across cohorts. (E) Pearson correlation analysis between pTSR and cTSR, indicating strong patient-level concordance. (F) Bland-Altman analysis assessing agreement between pTSR and cTSR. (G) Association between pTSR and overall survival in the JLU cohort, demonstrating a non-linear relationship consistent with that observed for cTSR. Together, these analyses establish the reliability of automated TSR quantification and its suitability for downstream survival modelling. ACC, accuracy; AI, artificial intelligence; AP, average precision; AUC, area under the receiver operating characteristic curve; cTSR, clinician-assessed tumour–stroma ratio; JLU, First Hospital of Jilin University; pTSR, AI-predicted tumour–stroma ratio; ROC, receiver operating characteristic; TCGA, The Cancer Genome Atlas; TSR, tumour–stroma ratio; WSI, whole-slide image.

$P$  for non-linear=0.050; figure 4G). A strong correlation was observed between the alternative (excluding necrosis) and original TSR measurements ( $r=0.764$ ,  $p<0.001$ ; online supplemental Fig. S2A). Sensitivity analyses using this alternative TSR formulation further reproduced the non-linear association with OS for both cTSR and pTSR in the JLU cohort (online supplemental Fig. S2), supporting the robustness of the primary findings. This concordance not only confirms the reliability of AI-derived TSR but also provides independent validation of the non-linear relationship between TSR and prognosis.

#### Performance, interpretability and non-linear modelling capacity of TSR-integrated multimodal prognostic models

We systematically compared the prognostic performance of seven models, including two traditional Cox regression models ( $\text{COX}_{\text{TNM}}$ ,  $\text{COX}_{\text{TSR+TNM+Age+Sex}}$ ) and five DL-based models ( $\text{DL}_{\text{WSI}}$ ,  $\text{DL}_{\text{WSI+TSR}}$ ,  $\text{DL}_{\text{WSI+TSR+TNM}}$ ,  $\text{DL}_{\text{WSI+TSR+TNM+Sex+Age}}$ ,  $\text{DL}_{\text{WSI+TSR+Extended Clinical}}$ ), across the JLU and TCGA cohorts (figure 5A–C). Ablation analyses showed that adding TSR to  $\text{DL}_{\text{WSI+TNM+Sex+Age}}$  and  $\text{DL}_{\text{WSI+Extended Clinical}}$  improved C-index and time-dependent AUCs (12-month, 36-month and 60-month) across cohorts. Addition of clinical variables also enhanced performance compared with WSI-only models (figure 5A, B; online supplemental Table S4; online supplemental Fig. S3A).

In the JLU internal test setting, traditional Cox models exhibited relatively limited discriminative ability, with a 60-month AUC of 0.635 for the  $\text{COX}_{\text{TNM}}$  model and 0.655 for the  $\text{COX}_{\text{TSR+TNM+Age+Sex}}$  model. In contrast, DL-based models consistently demonstrated superior discrimination, with stepwise performance gains observed as TSR and clinical information were progressively incorporated. Specifically, the 60-month AUC increased from 0.655 for the  $\text{DL}_{\text{WSI}}$  model to 0.738 for the  $\text{DL}_{\text{WSI+TSR}}$  model, and further to 0.803 and 0.802 for the  $\text{DL}_{\text{WSI+TSR+TNM}}$  and  $\text{DL}_{\text{WSI+TSR+TNM+Age+Sex}}$  models, respectively (figure 5A; online supplemental Table S4).

Model performance was further quantified using C-index under a fivefold cross-validation framework, including the JLU training framework, JLU internal testing and the TCGA external validation cohort (figure 5B). In the JLU cohort, DL-based models exhibited a progressive increase in C-index with incremental multimodal integration, whereas traditional Cox models showed comparatively modest performance (figure 5B; online supplemental Table S4). Notably, the  $\text{DL}_{\text{WSI+TSR+Extended Clinical}}$  model achieved the highest discriminative performance, with C-indices of 0.813 in fivefold cross-validated training and 0.757 in fivefold cross-validated testing.

Consistent performance trends were observed in the TCGA external validation cohort. Among all models, the  $\text{DL}_{\text{WSI+TSR+TNM+Age+Sex}}$  model achieved the highest 60-month AUC and C-index, outperforming both traditional Cox models and structurally simpler DL models (figure 5A, B; online supplemental Table S4), further supporting the robustness and generalisability of the multimodal framework across independent cohorts.

Time-dependent ROC analyses demonstrated that the multimodal models maintained stable discriminative performance at both 12-month and 36-month follow-up time points in the JLU and TCGA cohorts (online supplemental Fig. S2A). Time-dependent calibration analyses at 12, 36 and 60 months were performed for the  $\text{DL}_{\text{WSI+Extended Clinical}}$  multimodal model in the JLU cohort. The calibration curves showed good agreement between predicted and observed survival probabilities across time points and cohorts, indicating adequate calibration of the model (online supplemental Fig. S3B). In addition, DCA

showed that, compared with the  $\text{COX}_{\text{TNM}}$  model,  $\text{DL}_{\text{WSI+Extended Clinical}}$  multimodal models provided greater net clinical benefit across a wide range of threshold probabilities, with particularly pronounced advantages within clinically relevant threshold ranges (figure 5C). Consistent net benefit patterns were also observed in time-specific analyses at 1 and 5 years in the JLU cohort (online supplemental Fig. S3C), further supporting the robustness and potential clinical utility of TSR-integrated multimodal models.

Collectively, these results indicate that while cTSR confers incremental prognostic value within conventional Cox regression models, its integration into a DL-driven multimodal framework yields the most substantial and clinically meaningful improvement in prognostic performance.

To enhance model interpretability, SHAP analysis was conducted to quantify the contributions of clinical variables and TSR-related features within the multimodal framework (figure 5D). In the JLU internal test set, TNM stage remained the most influential predictor, whereas TSR consistently ranked among the top contributing features in both the  $\text{DL}_{\text{WSI+TSR+TNM+Age+Sex}}$  and  $\text{DL}_{\text{WSI+TSR+Extended Clinical}}$  models. A similar pattern of feature importance was reproduced in the TCGA external validation cohort using harmonised clinical variables, further supporting the robustness and cross-cohort generalisability of TSR as a key prognostic determinant.

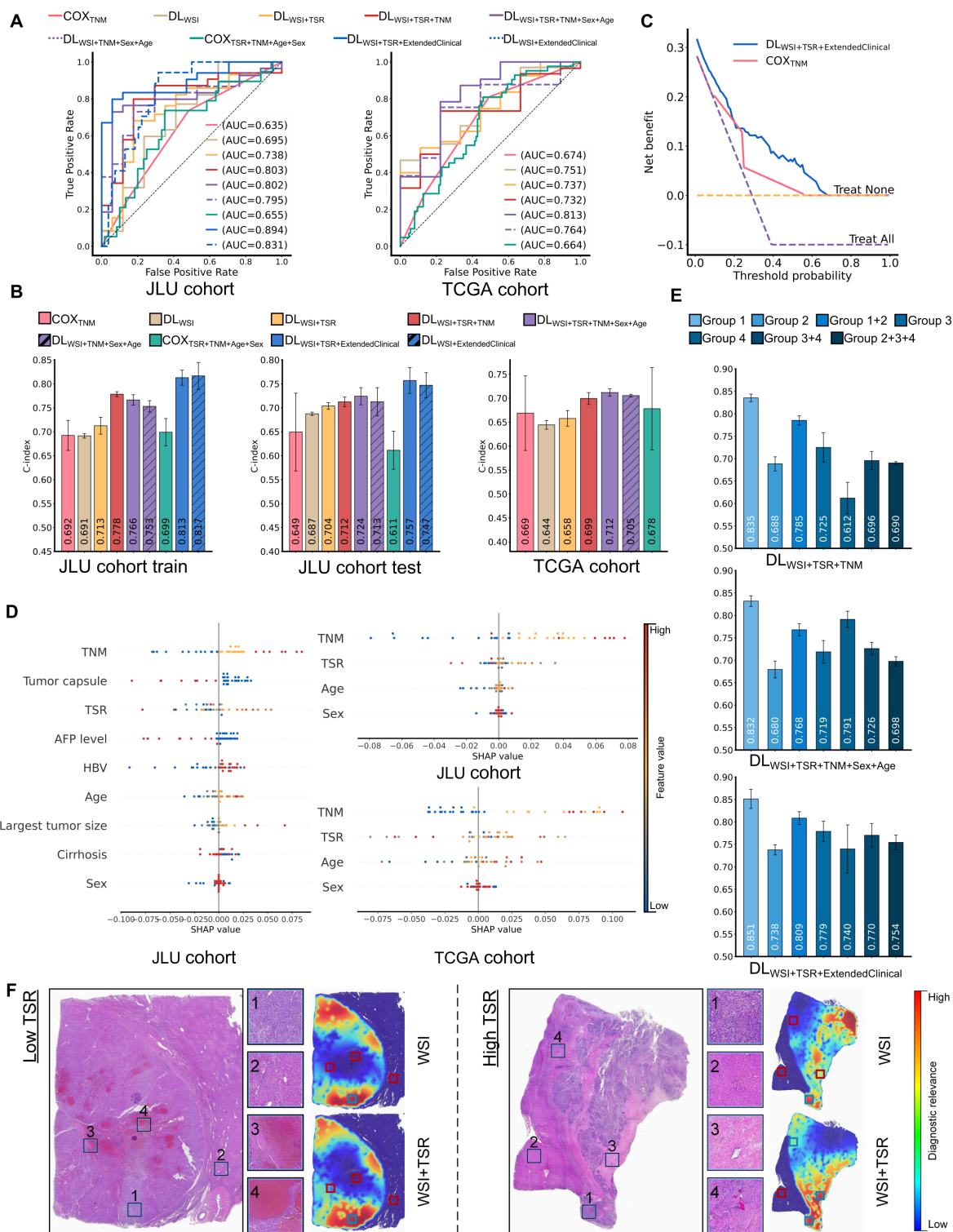
To further explore the potential non-linear association between TSR and prognosis, model performance was evaluated across RCS-defined TSR subgroups and subgroup combinations (figure 5E; online supplemental Fig. S3D). In both the JLU and TCGA cohorts, multimodal models incorporating TSR exhibited heterogeneous and non-monotonic performance patterns across individual TSR subgroups and their combinations, consistent with the non-linear association between TSR and clinical outcomes suggested by RCS analysis.

To visually illustrate the spatial impact of incorporating TSR into the multimodal framework, representative low-TSR and high-TSR cases were selected for visualisation using WSI-level attention heatmaps. Tumour regions showed consistently high activation in both the  $\text{DL}_{\text{WSI}}$  and  $\text{DL}_{\text{WSI+TSR}}$  models, whereas stromal regions enriched in stromal components were preferentially highlighted as high-importance regions only after TSR integration. These findings indicate that incorporation of TSR effectively recalibrated model attention toward biologically relevant stromal compartments (figure 5F).

#### Biological characterisation of TSR-associated non-linear prognostic patterns

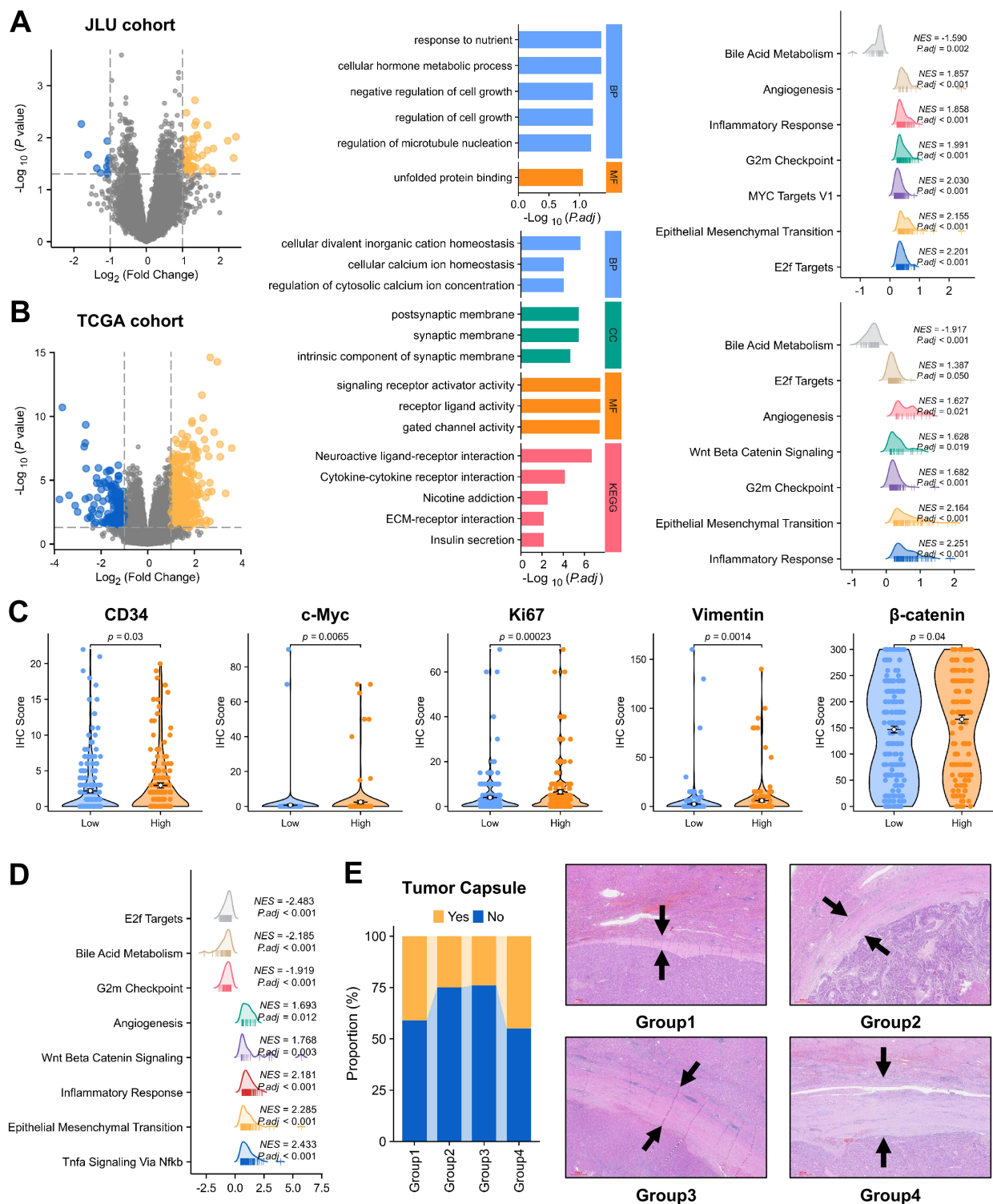
To investigate the biological correlates of the non-linear association between TSR and prognosis, transcriptomic analyses were performed in both cohorts. RNA data were available for 32 of 371 patients in the JLU cohort and 159 patients in the TCGA cohort. Based on the RCS-derived cut-off value (0.188), patients were classified into a low-TSR group (Group 1,  $n=22$ ) and a high-TSR group (Groups 2–4 combined,  $n=10$ ).

In the JLU cohort, differential gene expression analysis revealed distinct transcriptional profiles between low-TSR and high-TSR groups (figure 6A). Functional enrichment analyses demonstrated that genes upregulated in the high-TSR group were primarily associated with cell cycle regulation, proliferative signalling, angiogenesis, inflammatory response and epithelial–mesenchymal transition (EMT), as supported by Gene Ontology, Kyoto Encyclopedia of Genes and Genomes and GSEA results (figure 6A, online supplemental Fig. S4A). Consistent findings



**Figure 5** TSR-integrated multimodal prognostic models: performance comparison with conventional approaches, clinical utility and ablation-demonstrated incremental value and interpretability of TSR. (A–B) Comparative discrimination of conventional Cox models, TSR-integrated multimodal models and corresponding ablation models across the internal (JLU) and external (TCGA) cohorts. (C) Decision curve analysis demonstrating net clinical benefit of TSR integration across clinically relevant threshold probabilities. (D) SHAP analysis showing the relative contribution of TSR and clinical features in multimodal models. (E) Model discrimination across RCS-defined TSR strata, illustrating heterogeneous prognostic effects across risk groups. (F) Representative WSI attention heatmaps comparing stromal-region activation before and after TSR integration. Together, these panels demonstrate that TSR enhances risk stratification within a multimodal framework while preserving clinical interpretability. AFP, alpha-fetoprotein; AUC, area under the curve; C-index, concordance index; Cox, Cox proportional hazards model; DL, deep learning; JLU, First Hospital of Jilin University; RCS, restricted cubic spline; SHAP, Shapley additive explanations; TCGA, The Cancer Genome Atlas; TNM, tumour, node, metastasis; TSR, tumour–stroma ratio; WSI, whole-slide image.





**Figure 6** Transcriptomic and histopathological characterisation of TSR-associated non-linear prognostic patterns. (A) Differential gene expression and functional enrichment analyses comparing low-TSR (Group 1) and high-TSR (Groups 2–4) subgroups in the JLU cohort (n=32). (B) Differential gene expression and functional enrichment analyses comparing low-TSR (Group 1) and high-TSR (Groups 2–4) subgroups in the TCGA cohort (n=159). (C) IHC analyses in the entire JLU cohort (n=371) comparing protein expression between low-TSR and high-TSR groups. (D) Gene set enrichment analysis comparing Group 2 and Group 3, located on opposite sides of the RCS-derived risk peak within the HR >1 region of the JLU cohort. (E) Distribution of tumour capsule presence across RCS-defined cTSR subgroups. BP, biological process; CC, cellular components; cTSR, clinician-assessed tumour–stroma ratio; IHC, immunohistochemistry; JLU, First Hospital of Jilin University; KEGG, Kyoto Encyclopedia of Genes and Genomes; MF, molecular function; NES, normalised enrichment score; RCS, restricted cubic spline; TCGA, The Cancer Genome Atlas; TSR, tumour–stroma ratio.

were observed in the TCGA cohort, where high-TSR tumours similarly exhibited enrichment of stromal-associated oncogenic programmes, including EMT, inflammatory response,

angiogenesis, cell cycle progression and WNT/β-catenin signalling, supporting the reproducibility of TSR-associated biological features across independent cohorts (figure 6B, online



supplemental Fig. S4B). The above conclusions were further supported using the 29 Fges defined by Bagaev *et al* for micro-environment subtype classification<sup>28</sup> (online supplemental Fig. S4C).

Protein-level using immunohistochemistry (IHC) in the entire JLU cohort (n=371) showed significantly increased expression of markers related to proliferation (Ki-67), oncogenic signalling (c-Myc and  $\beta$ -catenin), angiogenesis (CD34) and mesenchymal phenotype (Vimentin) in the high-TSR group compared with the low-TSR group (figure 6C).

To further characterise biological heterogeneity within the RCS-defined HR >1 region, transcriptomic differences between Group 2 (n=6) and Group 3 (n=4) were examined (online supplemental Fig. S4D). Group 3 was characterised by attenuated proliferation-related pathways (E2F targets and G2/M checkpoint) alongside enrichment of inflammation-associated and immune-associated signalling programmes, including inflammatory response and TNF $\alpha$ -NF- $\kappa$ B signalling (figure 6D, online supplemental Fig. S4E-F). Consistent with the findings in the JLU cohort, orthogonal analysis in the TCGA cohort demonstrated similar transcriptomic patterns between Group 2 and Group 3, characterised by attenuation of proliferation-related programmes and relative enrichment of immune-associated and inflammation-associated pathways (online supplemental Fig. S5G). These findings suggest that the declining hazard beyond the risk peak may be associated with differences in transcriptomic patterns, particularly involving proliferation-related and immune-related pathways.

To explore the pathological basis of the favourable prognosis observed in the ultra-high TSR range (>0.392), we examined capsule status across RCS-defined subgroups. Capsule presence was more frequent in Group 4 (45%) than in Groups 1–3 (24%–41%), suggesting that the favourable prognosis observed in the ultra-high TSR range may be partly attributable to capsular containment rather than intratumoral stromal accumulation (figure 6E).

## DISCUSSION

While TSR is a recognised prognostic factor, its clinical utility has been limited by conflicting quantification methods and linear assumptions.<sup>14 29 30</sup> Here, we moved beyond traditional scalar measurements by encoding TSR as a high-dimensional feature within a multimodal framework, unveiling an inverted U-shaped relationship that may help explain prior prognostic inconsistencies. Notably, application of the RCS-derived threshold (0.188) extended effective risk stratification from early-stage to advanced-stage populations (MVI-positive, TNM stage II–III), suggesting its potential to complement existing staging systems across the broader disease spectrum. Our model achieved superior stratification in both JLU and TCGA cohorts, with biological interpretability validated through transcriptomic and IHC analyses. The incremental improvement observed in ablation analyses suggests that TSR provides non-redundant prognostic information beyond standard clinical factors, supporting its integration into postoperative risk stratification. We now state that the integration of TSR offers a finer granularity of risk assessment, particularly for patients where TNM staging alone may not fully capture the risk of OS.

Using manual delineation as the reference standard, we further supported the prognostic value of TSR while revealing a distinct non-linear relationship that contrasts with prior linear assumptions.<sup>9 10 13 16</sup> Specifically, we identified a risk initiation threshold at 0.188, which is significantly lower than the

conventional 0.5 cut-off used in most previous studies.<sup>10 13</sup> This finding aligns with our previous work on HCC gross classification<sup>31</sup>: the most aggressive phenotype—type IV—contained only 28.3% low-TSR cases, compared with 54.5%–74.1% in types I–III. Beyond this threshold, the high-risk phenotype is associated with transcriptional programmes related to active tumour proliferation (MYC, E2F), stromal remodelling (EMT, angiogenesis) and TME crosstalk (IL6 and WNT signalling). The HR increases progressively until plateauing around 0.268, then reverses in the ultra-high range, suggesting a potential attenuated risk pattern beyond 0.392.

Morphologically, cases in this ultra-high TSR range (>0.392) are characterised by a thick peritumorous capsule rather than abundant intratumoral stroma, suggesting that capsular stroma may function as a distinct biological entity from intratumoral stroma.<sup>32 33</sup> Dissecting the spatial heterogeneity of stromal components—specifically distinguishing peritumorous capsule from intratumoral matrix—represents a promising direction for future HCC research.

Historically, TSR assessment has been limited by inconsistencies in region of interest selection. Unlike previous studies that relied on stroma-rich ‘hotspots’ or random selection,<sup>14</sup> we adapted our approach to the large tumour volume typical of HCC using a standardised seven-point pathological sampling framework to select digital images that represent the tumour’s global features for subsequent modelling. The reproducibility of TSR-associated prognostic patterns across cohorts and across TSR definitions (cTSR and pTSR) provides indirect support for the robustness of this representative-section strategy. However, regarding quantification, manual visual estimation is inherently subjective and poorly reproducible.<sup>34</sup> While supervised segmentation networks (eg, UNet++) automate this process,<sup>35 36</sup> they remain bound to a rigid ‘pixel-counting’ paradigm. These methods demand expensive pixel-level annotations yet remain fragile to stain variations and artefacts, limiting their real-world utility.<sup>37</sup> Fundamentally, treating TSR as a simple scalar may overlook the multidimensional nature of the stroma—texture, density and spatial arrangement—and may impose a linear interpretation onto a non-linear biological reality.

Our study represents a methodological evolution to bridge this gap. The inverted U-shaped non-linear relationship between TSR and prognosis suggests that TSR functions as a proxy for TME heterogeneity, requiring high-dimensional decoding rather than simple quantification. Accordingly, we moved beyond segmentation by deploying a pipeline that couples a pathology-specific foundation model (Conch v1.5) with a lightweight MLP classifier. Conch v1.5, pretrained on millions of histopathology images,<sup>20</sup> extracts tissue patches as high-dimensional semantic tokens rather than reductionist scalar ratios.<sup>38</sup> This architectural shift enables our framework to decode complex biological patterns embedded within distinct tissue phenotypes.

By developing a ‘Token-Guided Multimodal Fusion’ architecture based on Titan,<sup>21</sup> we integrated WSI, TSR and clinical variable as high-dimensional tokens within a unified framework. This approach enriched feature representation and preserved cross-modal associations, achieving improved prognostic accuracy (AUC 0.802 and 0.813) and cross-cohort generalisability compared with unimodal approaches. DCA further demonstrated greater net clinical benefit across threshold probabilities of 0.2–0.6. In addition, the good agreement between predicted and observed survival probabilities across 12-month, 36-month and 60-month horizons supports the reliability of the multimodal model for postoperative risk assessment. Notably, SHAP analyses ranked TSR among the top prognostic contributors.

Beyond performance gains, encoding TSR as a high-dimensional token enabled the model to autonomously capture the inverted U-shaped pattern, highlighting the critical importance of preserving information richness in AI modelling.

Regarding interpretability, we further identified a critical limitation in  $DL_{WSI}$  survival modelling: a 'high-cellularity bias' (figure 5F), where DL algorithms preferentially focused on dense tumour regions while treating stroma as background noise.<sup>37 39</sup> The introduction of the TSR token effectively recalibrated the network's attention mechanism to explicitly recognise stromal components and the tumour–stroma interface. In this context, the TSR token functions as a 'semantic anchor', synergising human domain knowledge with data-driven AI capabilities to capture complex TME interactions. This fusion bridges the gap between data-driven pattern recognition and human domain-informed biological reasoning.

Furthermore, the use of the Conch v1.5 foundation model extracts robust, domain-invariant morphological semantics, achieving a near-perfect correlation with pathologist assessments ( $R^2 > 0.9$ ) and demonstrated superior cross-ancestry stability across both Asian and Western cohorts without motivating extensive manual annotation.

Clinically, our findings highlight the value of incorporating TME features into pathological assessment, alongside conventional tumour-centric metrics such as tumour size, number and vascular invasion. By quantifying stromal proportion, TSR-informed stratification provides a stromal dimension that complements existing staging systems, which remain central to clinical decision-making in HCC, particularly in early stages. Importantly, this study provides insight into how stromal proportion shapes tumour behaviour and patient risk. The AI-based framework supports standardised and reproducible evaluation of TSR, thereby enabling consistent and clinically interpretable characterisation of TSR-associated risk patterns. Such stratification may support postoperative risk assessment and surveillance, linking quantitative modelling with clinically meaningful interpretation rather than being limited to individualised survival prediction. Future studies are anticipated to further investigate the potential predictive relevance of TSR-associated biological phenotypes in relation to treatment response across therapeutic modalities.

This study has several limitations. First, although representative sections were selected based on a standardised seven-point sampling protocol to account for spatial heterogeneity in HCC, a single section may not fully capture intratumoral heterogeneity. Future studies incorporating multiregional sampling or spatially resolved analyses are warranted. Second, validation in more diverse populations and clinical settings is warranted. Third, while we provided preliminary mechanistic insights through transcriptomic and histopathological analyses, the biological basis of the non-linear TSR–prognosis relationship requires further investigation; development of experimental models simulating varying TSR proportions may help elucidate the underlying mechanisms. Finally, the interpretability of DL-derived features, particularly how the model captures non-linear TSR dynamics, remains an evolving area that warrants continued exploration. Future research should address these limitations and further refine the proposed AI framework.

In summary, by synergising foundation model-driven semantics with a novel multimodal fusion architecture, our study transforms TSR from a static scalar to a dynamic, high-dimensional biomarker, effectively reshaping the prognostic landscape of the TME. Crucially, our framework integrates human priors to overcome the 'high-cellularity bias' typical of DL models,

autonomously capturing non-linear prognostic features while providing morphological interpretation rooted in stromal spatial heterogeneity. Clinically, this approach offers a standardised, annotation-free tool for TSR-based risk stratification, enabling consistent cross-centre implementation and providing a stromal dimension to complement existing TNM staging. Ultimately, our study implicates that the future of computational pathology may lie not only in simple quantification, but in the deep semantic fusion of human domain knowledge with high-dimensional AI reasoning as well.

#### Author affiliations

<sup>1</sup>Department of Hepatobiliary and Pancreatic Surgery, General Surgery Center, First Hospital of Jilin University, Changchun, Jilin, China

<sup>2</sup>China-Singapore Belt and Road Joint Laboratory on Liver Disease Research, First Hospital of Jilin University, Changchun, Jilin, China

<sup>3</sup>The Image Processing Center, School of Astronautics, Beihang University, Beijing, China

<sup>4</sup>Department of Pathology, First Hospital of Jilin University, Changchun, Jilin, China

<sup>5</sup>Division of Hepatobiliary & Pancreatic Surgery, Department of Surgery, National University Hospital Singapore, Singapore

<sup>6</sup>School of Engineering Medicine, Beijing Advanced Innovation Center on Biomedical Engineering, Beihang University, Beijing, China

**Acknowledgements** We would like to thank the Department of Biobank, Division of Clinical Research, First Hospital of Jilin University, for constructing the tissue microarrays. We also acknowledge Professor Yanhua Wu for her support in statistical analysis. Some published data used in this study were generated by the TCGA Research Network (<https://www.cancer.gov/tcga>). Some figures were created using BioRender.com.

**Contributors** Conception: GYL, ZQF, FYX and YSZ. Study design: ZQF, HYH and YSZ. Administrative support: GYL, ZQF and FYX. Data collection and acquisition: HYH, LMQ, XDS, FW, PZ, MSJ, LG, YGC, WQ, MW, XJS, JFY and YH. Data analysis: KW, HYH, MYL, CHH and YXZ. Manuscript preparation: ZQF, HYH, KW, CHH and MYL. Critical revision: ZQF, GYL, FYX, YSZ and AWCK. Final approval of manuscript: All authors. HYH, KW and LMQ contributed equally to this work. GYL acts as the guarantor for this article. AI-assisted tools (Claude 4.5) were used for language editing and manuscript polishing. The authors reviewed and took full responsibility for the content.

**Funding** This work was supported by the National Key Research and Development Program of China (No 2024YFE0213800), the Natural Science Foundation of China (No 82473447), the Jilin Provincial Science and Technology Development Program (No 202502001GH) and the Fundamental Research Funds for the Central Universities.

**Competing interests** None declared.

**Patient and public involvement** Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

**Patient consent for publication** Not applicable.

**Ethics approval** This study was approved by the Institutional Review Board of the First Hospital of Jilin University (approval number: 2025-042) and conducted in accordance with the Declaration of Helsinki. Informed consent was waived owing to the retrospective design. TCGA data use complied with its data use agreement.

**Provenance and peer review** Not commissioned; externally peer reviewed.

**Data availability statement** Data are available upon reasonable request. The JLU cohort data are available from the corresponding author upon reasonable request. TCGA data are publicly available through the GDC Data Portal.

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## ORCID iDs

Feng Wei <https://orcid.org/0000-0001-8451-4516>Guo-Yue Lv <https://orcid.org/0000-0003-4656-4955>

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